

DD_OSR-05 - Inventory Audit

Version: v1.0 **Bundle:** 08_osr_equipment **Status:** Draft for review **Primary owner:** OSR / warehouse operations team

Related DDs: DD_OSR-01, DD_OSR-INSTANCE-01, DD_OSR-02, DD_EM-02, DD_EM-CFG-03, DD_EM-CFG-04, DD_ICN-01, DD_REP-01, FOUNDATION_DROOLS

MVP simplification baseline: implement stock and equipment instance basics before full inventory audit. OSR-05 can start with manual count import, discrepancy list, and approval/write-off routing; advanced mobile count campaigns can be phased later.

1. Purpose

OSR-05 defines physical inventory audit, discrepancy investigation, stock adjustment, and write-off governance for SOPHIX equipment stock.

It covers central warehouse counts, contractor van counts, serialized equipment checks, discrepancy review, approved stock adjustments, and write-offs for lost, stolen, damaged, obsolete, or unrecoverable equipment.

Use OSR-05 when:

- warehouse team performs a monthly/quarterly stock count;
- contractor van inventory must be checked;
- system stock differs from physical stock;
- serial device physically exists but system says it is elsewhere;
- system shows equipment but staff cannot find it;
- inventory discrepancy requires investigation;
- approved adjustment/write-off must update OSR-01 and OSR-INSTANCE.

OSR-05 does not own stock balances or device lifecycle. It investigates differences and calls OSR-01/OSR-INSTANCE to apply approved corrections.

2. Scope And Boundaries

Capability	Owner
Inventory audit plan, count tasks, count results, discrepancy investigation	OSR-05
SKU quantity balance and adjustment movement	OSR-01
Serialized equipment current state and lifecycle transition	OSR-INSTANCE-01
Procurement receipt source records	OSR-02
Contractor/team location ownership	EM-02
Audit permissions	EM-CFG-03
Configurable write-off/adjustment approval	EM-CFG-04
Configurable discrepancy severity/routing	Drools
Internal audit notifications	ICN-01
Audit reporting	REP-01

Boundary rules:

- OSR-05 must not directly edit stock_balance.
- Quantity corrections are applied by OSR-01 stock adjustment movements.
- Serialized write-offs/state corrections are applied by OSR-INSTANCE lifecycle events.
- EM-CFG-04 owns approval policies and tasks. OSR-05 stores approval references and applies final outcome.
- Camunda is used only when EM-CFG-04 selected policy mode is BPMN.

3. Process Overview

```
flowchart LR
    A["Create audit plan"] --> B["Generate count tasks"]
    B --> C["Counter records physical quantities/serials"]
    C --> D["System compares expected vs counted"]
    D --> E["Discrepancy opened"]
    E --> F["Investigation"]
    F --> G["Approval if required"]
    G --> H["Apply OSR-01 adjustment"]
    H --> I["Apply OSR-INSTANCE lifecycle correction/write-off"]
    I --> J["Audit closed"]
```

4. Status Models

4.1 Audit Plan Status

Status	Meaning
DRAFT	Audit plan being prepared.
SCHEDULED	Plan approved/scheduled.
IN_PROGRESS	Counting started.
RECONCILING	Counts complete; differences being reviewed.
PENDING_APPROVAL	Adjustment/write-off approval open.
CLOSED	Audit completed.
CANCELLED	Audit cancelled.

4.2 Discrepancy Status

Status	Meaning
OPEN	Difference detected.
INVESTIGATING	Assigned to reviewer.
PENDING_APPROVAL	Resolution needs EM-CFG-04 approval.
APPROVED	Resolution approved.
APPLIED	OSR correction applied.
REJECTED	Proposed resolution rejected.
CLOSED	Fully handled.

5. Data Model

5.1 inventory_audit_plan

What it is for: stores one planned physical inventory audit.

When to use it: create a plan for central warehouse count, contractor van count, cycle count, surprise audit, or investigation audit.

Sample meaning: iap-001 is a monthly WIK Nairobi central warehouse count scheduled for July 31.

Column	Type	Purpose	Sample
audit_plan_id	TEXT PK	Audit plan id.	iap-001
operator_code	TEXT	Tenant/operator.	WIK
audit_type	TEXT	CYCLE_COUNT, FULL_COUNT, SPOT_CHECK, INVESTIGATION.	CYCLE_COUNT
location_id	TEXT	OSR-01 stock location.	WIK-CENTRAL-NRB
contractor_id	TEXT NULL	Contractor if van/location belongs to contractor.	contractor_ke_nairobi_a
scheduled_start_at	TIMESTAMP	Planned start.	2026-07-31T08:00:00Z

Column	Type	Purpose	Sample
scheduled_end_at	TIMESTAMP	Planned end.	2026-07-31T17:00:00Z
status	TEXT	Audit plan status.	SCHEDULED
created_by_user_id	TEXT	Creator.	usr-inventory-manager
created_at	TIMESTAMP	Creation timestamp.	2026-07-20T10:00:00Z
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- (operator_code, location_id, status).
- (operator_code, scheduled_start_at).

5.2 inventory_count_task

What it is for: stores count assignments generated from an audit plan.

When to use it: assign one or more staff/contractors to count specific SKUs, zones, bins, or serialized devices.

Sample meaning: ict-001 asks warehouse user usr-counter-001 to count Huawei ONTs in bin A-01.

Column	Type	Purpose	Sample
count_task_id	TEXT PK	Count task id.	ict-001
audit_plan_id	TEXT	Parent audit plan.	iap-001
location_id	TEXT	Count location.	WIK-CENTRAL-NRB
sku_id	TEXT NULL	SKU to count; null means all assigned scope.	WIK-ONT-HUAWEI-EG8145V5
bin_code	TEXT NULL	Physical bin/shelf.	A-01
assigned_to_user_id	TEXT	Counter.	usr-counter-001
status	TEXT	ASSIGNED, IN_PROGRESS, SUBMITTED, RECOUNT_REQUIRED, ACCEPTED.	SUBMITTED
started_at	TIMESTAMP NULL	Start time.	2026-07-31T08:30:00Z
submitted_at	TIMESTAMP NULL	Submission time.	2026-07-31T09:10:00Z

Recommended indexes:

- (audit_plan_id, status).
- (assigned_to_user_id, status).

5.3 inventory_count_line

What it is for: stores counted quantity and serials for one count task line.

When to use it: each counted SKU/bin or serial capture submits a count line.

Sample meaning: icl-001 says the counter found 97 Huawei ONTs while system expected 100.

Column	Type	Purpose	Sample
count_line_id	TEXT PK	Count line id.	icl-001
count_task_id	TEXT	Parent task.	ict-001
sku_id	TEXT	SKU counted.	WIK-ONT-HUAWEI-EG8145V5
expected_qty	INT	System quantity from OSR-01 at count freeze.	100
counted_qty	INT	Physical quantity counted.	97
variance_qty	INT	Counted minus expected.	-3
serials_counted_json	JSONB NULL	Serial list for serialized SKU.	["48575443ABCD1234"]
counter_comment	TEXT NULL	Counter note.	Three units missing from bin.
submitted_at	TIMESTAMP	Submit timestamp.	2026-07-31T09:10:00Z

Recommended indexes:

- (count_task_id).
- (sku_id).
- (variance_qty).

5.4 inventory_discrepancy

What it is for: stores detected variance requiring review.

When to use it: count variance is not zero, serial mismatch exists, location mismatch exists, or count quality fails.

Sample meaning: idisc-001 is a shortage of 3 Huawei ONTs at WIK central warehouse.

Column	Type	Purpose	Sample
discrepancy_id	TEXT PK	Discrepancy id.	idisc-001
operator_code	TEXT	Tenant/operator.	WIK
audit_plan_id	TEXT	Audit plan.	iap-001
count_line_id	TEXT NULL	Source count line.	icl-001
location_id	TEXT	Location.	WIK-CENTRAL-NRB
sku_id	TEXT	SKU.	WIK-ONT-HUAWEI-EG8145V5
discrepancy_type	TEXT	SHORTAGE, OVERAGE, SERIAL_MISMATCH, LOCATION_MISMATCH, DAMAGED_FOUND.	SHORTAGE
quantity_delta	INT	Counted minus expected.	-3
estimated_value	NUMERIC(14,2) NULL	Estimated financial impact.	18000.00
severity	TEXT	LOW, MEDIUM, HIGH, CRITICAL.	HIGH
status	TEXT	Discrepancy status.	INVESTIGATING
assigned_to_user_id	TEXT NULL	Investigator.	usr-inventory-supervisor
created_at	TIMESTAMP	Creation timestamp.	2026-07-31T09:15:00Z
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- (operator_code, status, created_at DESC).
- (audit_plan_id, status).
- (location_id, sku_id).

5.5 inventory_resolution

What it is for: stores proposed and applied discrepancy resolution.

When to use it: investigator decides whether to recount, accept variance, adjust stock, write off, transfer correction, or mark found.

Sample meaning: ires-001 proposes to write off 3 missing ONTs after investigation.

Column	Type	Purpose	Sample
resolution_id	TEXT PK	Resolution id.	ires-001
discrepancy_id	TEXT	Discrepancy id.	idisc-001
resolution_action	TEXT	RECOUNT, ADJUST_STOCK, WRITE_OFF, FOUND_AND_CORRECT_LOCATION, NO_ACTION.	WRITE_OFF
proposed_by_user_id	TEXT	Proposer.	usr-inventory-supervisor
approval_request_id	TEXT NULL	EM-CFG-04 request if required.	apr-wof-001
approval_mode	TEXT NULL	EM-CFG-04 mode.	MULTI_STEP
camunda_process_instance_id	TEXT NULL	BPMN instance if used.	cam-wof-001

Column	Type	Purpose	Sample
status	TEXT	PROPOSED, PENDING_APPROVAL, APPROVED, REJECTED, APPLIED.	PENDING_APPROVAL
comment	TEXT NULL	Resolution note.	Not found after recount and camera review.
created_at	TIMESTAMP	Creation timestamp.	2026-07-31T11:00:00Z
applied_at	TIMESTAMP NULL	Apply timestamp.	null

Recommended indexes:

- (discrepancy_id, status).
- (approval_request_id).

5.6 inventory_write_off_item

What it is for: stores item-level write-off detail for approved write-off resolutions.

When to use it: resolution action is WRITE_OFF, especially when serialized instances are involved.

Sample meaning: iwoi-001 writes off missing serial 48575443ABCD1234.

Column	Type	Purpose	Sample
write_off_item_id	TEXT PK	Write-off item id.	iwoi-001
resolution_id	TEXT	Resolution id.	ires-001
sku_id	TEXT	SKU.	WIK-ONT-HUAWEI-EG8145V5
quantity	INT	Quantity written off.	1
instance_id	TEXT NULL	OSR-INSTANCE id if serialized.	inst-ont-001
serial_number	TEXT NULL	Serial if known.	48575443ABCD1234
reason_code	TEXT	LOST, STOLEN, DAMAGED, OBSOLETE, UNRECOVERABLE.	LOST
estimated_value	NUMERIC(14,2) NULL	Estimated item value.	6000.00

Recommended indexes:

- (resolution_id).
- (instance_id).
- (sku_id).

6. APIs

6.1 Create Audit Plan

```
POST /api/inventory-audits
Idempotency-Key: audit-wik-central-20260731
Content-Type: application/json

{
  "operatorCode": "WIK",
  "auditType": "CYCLE_COUNT",
  "locationId": "WIK-CENTRAL-NRB",
  "scheduledStartAt": "2026-07-31T08:00:00Z",
  "scheduledEndAt": "2026-07-31T17:00:00Z",
  "scope": {
    "skuIds": ["WIK-ONT-HUAWEI-EG8145V5"],
    "binCodes": ["A-01", "A-02"]
  }
}
```

6.2 Generate Count Tasks

```
POST /api/inventory-audits/{audit_plan_id}/generate-tasks
Content-Type: application/json
```

Creates inventory_count_task rows from scope and assigned counters.

6.3 Submit Count

POST /api/inventory-count-tasks/{count_task_id}/submit
Content-Type: application/json

```
{
  "lines": [
    {
      "skuId": "WIK-ONT-HUAWEI-EG8145V5",
      "countedQty": 97,
      "serialsCounted": ["48575443ABCD1234"],
      "counterComment": "Three units missing from bin."
    }
  ]
}
```

OSR-05 compares against frozen OSR-01 balance and OSR-INSTANCE expected serial list.

6.4 Propose Resolution

POST /api/inventory-discrepancies/{discrepancy_id}/resolutions
Content-Type: application/json

```
{
  "resolutionAction": "WRITE_OFF",
  "comment": "Not found after recount and camera review.",
  "items": [
    {
      "skuId": "WIK-ONT-HUAWEI-EG8145V5",
      "quantity": 3,
      "reasonCode": "LOST"
    }
  ]
}
```

OSR-05 evaluates whether approval is required through Drools/EM-CFG-04.

6.5 Approval Outcome Callback

POST /internal/inventory-resolutions/{resolution_id}/approval-outcome
Content-Type: application/json

```
{
  "approvalRequestId": "apr-wof-001",
  "outcome": "APPROVED",
  "decisionComment": "Approved after inventory investigation."
}
```

6.6 Apply Resolution

POST /api/inventory-resolutions/{resolution_id}/apply
Content-Type: application/json

For ADJUST_STOCK or WRITE_OFF, OSR-05 calls OSR-01:

POST /api/stock-movements
Idempotency-Key: inv-resolution-ires-001-stock
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "movementType": "ADJUSTMENT",
  "reasonCode": "INVENTORY_WRITE_OFF",
  "skuId": "WIK-ONT-HUAWEI-EG8145V5",
  "quantity": -3,
  "sourceLocationId": "WIK-CENTRAL-NRB",
  "originRefType": "INVENTORY_RESOLUTION",
  "originRefId": "ires-001",
  "actorUserId": "usr-inventory-supervisor"
}
```

For serialized item write-off, OSR-05 calls OSR-INSTANCE:

POST /api/equipment-instances/{instance_id}/lifecycle-events
Idempotency-Key: inv-writeoff-ires-001-inst-ont-001
Content-Type: application/json

```
{
  "toState": "DECOMMISSIONED",
  "reasonCode": "INVENTORY_WRITE_OFF_LOST",
  "originRefKind": "INVENTORY_RESOLUTION",
  "originRefId": "ires-001",
}
```

```

    "actorUserId": "usr-inventory-supervisor",
    "notes": "Approved write-off after physical count discrepancy."
  }

```

7. Business Rules

Rule	Description	Reason
R-OSR-05-01	Audit location must exist in OSR-01.	Valid count scope.
R-OSR-05-02	Audit expected quantity is frozen at task generation/start.	Stable variance calculation.
R-OSR-05-03	Count submitter cannot approve own discrepancy resolution unless EM-CFG-04 explicitly allows it.	Segregation of duties.
R-OSR-05-04	Non-zero variance creates discrepancy unless tolerance policy suppresses low-risk variance.	Audit traceability.
R-OSR-05-05	Stock adjustment/write-off must be approved through EM-CFG-04 when policy requires it.	Governance.
R-OSR-05-06	Camunda is used only when EM-CFG-04 policy mode is BPMN.	Operational simplicity.
R-OSR-05-07	Approved quantity correction must call OSR-01 movement API.	Stock ownership.
R-OSR-05-08	Serialized write-off must call OSR-INSTANCE lifecycle API.	Serial traceability.
R-OSR-05-09	Applied resolution must be idempotent.	Prevent double adjustment.
R-OSR-05-10	Audit closure requires all discrepancies closed/rejected/no-action.	Complete audit trail.

8. Rule Engine Guidance

Use Drools for configurable audit policy:

Rule file	Purpose
inventory-discrepancy-severity.drl	Assign severity based on value, quantity, SKU family, location, and contractor.
inventory-resolution-approval.drl	Decide EM-CFG-04 approval policy for write-off/adjustment.
inventory-recount-policy.drl	Decide when recount is mandatory before resolution.

Example:

```

package com.sophix.osr.inventory

rule "High value write off requires finance approval"
when
    $r : InventoryResolutionFact(action == "WRITE_OFF", estimatedValue >= 10000.00, currencyCode == "KES")
    $d : InventoryResolutionDecision()
then
    $d.setApprovalRequired(true);
    $d.setApprovalPolicyCode("INVENTORY_WRITE_OFF_FINANCE");
    $d.addReason("HIGH_VALUE_WRITE_OFF");
end

```

9. Events

Event	Topic	When emitted
InventoryAuditCreated	sophix.osr.inventory.audit-created	Audit plan created.
InventoryCountSubmitted	sophix.osr.inventory.count-submitted	Counter submits count.
InventoryDiscrepancyOpened	sophix.osr.inventory.discrepancy-opened	Variance detected.
InventoryResolutionApprovalRequired	sophix.osr.inventory.approval-required	EM-CFG-04 request started.
InventoryResolutionApplied	sophix.osr.inventory.resolution-applied	OSR correction applied.

Event	Topic	When emitted
InventoryAuditClosed	sophix.osr.inventory.audit-closed	Audit fully closed.

10. Frontend/API Consumers

Area	APIs/modules used
Backoffice inventory audit	OSR-05 audit, task, discrepancy, resolution APIs.
Contractor mobile app	Count task submit APIs for van audits if allowed.
Approval inbox	EM-CFG-04 task APIs.
Stock view	OSR-01 balances plus OSR-05 open discrepancy badges.
Serial lookup	OSR-INSTANCE details for serialized discrepancies.
Reporting	REP-01 consumes OSR-05 events.

11. Infrastructure Recommendations

For medium deployment, run OSR-05 inside the OSR/equipment API and shared OSR worker deployment.

Component	Recommendation
API	Add audit/discrepancy endpoints to OSR API.
Database	PostgreSQL OSR schema.
Workers	Shared OSR worker for outbox, reminders, and resolution retry.
Mobile offline	Contractor app can cache assigned count tasks and sync submissions with idempotency.
Approval	EM-CFG-04 shared service.
Rules	Drools/KIE for configurable audit routing.

AWS notes: use ECS/EKS OSR service, RDS PostgreSQL, S3 for audit evidence/photos/files, MSK/Kafka for events, and Secrets Manager for credentials.

12. Acceptance Criteria

- Audit plan/task/count/discrepancy/resolution tables exist.
- Audit count freezes expected OSR-01 quantities.
- Variance creates auditable discrepancy with severity.
- Resolution approvals use EM-CFG-04; OSR-05 does not hardcode approval steps.
- Approved stock correction calls OSR-01 movement API.
- Approved serialized write-off calls OSR-INSTANCE lifecycle API.
- Contractor mobile app can submit assigned counts where configured.
- Tests cover count variance, approval, idempotent apply, OSR-01 adjustment, and OSR-INSTANCE write-off.